The ER diagram represents the database structure of the Unbound platform, which is used to manage college fests, events, student registrations, teams, attendance, and certificates.

The diagram consists of multiple entities (tables) connected through relationships using primary keys and foreign keys.

The main goal of this database is to organize event-related information efficiently while maintaining data integrity and avoiding redundancy.

1. Colleges Entity

The Colleges table stores information about colleges participating in the system.

Attributes

- college_id (Primary Key) – unique identifier for each college
- name – name of the college
- location – city or region of the college
- description – additional information about the college
- created_at – timestamp when the record was created

Relationship

A college can have:

- multiple clubs
- multiple users
- multiple fests

So the relationship is:

College → Clubs (1:N)
College → Users (1:N)
College → Fests (1:N)

2. Users Entity

The Users table stores all system users.

Users may have different roles such as:

- Student
- Club Admin
- College Admin
- Super Admin

Attributes

- user_id (Primary Key) – unique user ID
- name – user name
- email – login email
- password – encrypted password
- role – user role in system
- college_id (Foreign Key) – college associated with the user
- created_at – account creation timestamp

Relationships

Users can:

- register for events
- join teams
- receive certificates
- have attendance records

Example:

User → Registrations
User → Attendance
User → Certificates

3. Clubs Entity

The Clubs table represents clubs inside a college that organize events.

Example:

- Coding Club
- Robotics Club
- Cultural Club

Attributes

- club_id (Primary Key)
- club_name – name of the club
- description – information about the club
- college_id (Foreign Key) – which college the club belongs to
- created_at – timestamp

Relationship

College (1) → (N) Clubs
Club (1) → (N) Events

Meaning one club can organize many events.

4. Fests Entity

The Fests table stores information about college festivals.

Example:

- Tech Fest
- Cultural Fest
- Annual Fest

Attributes

- fest_id (Primary Key)
- fest_name – name of the festival
- college_id (Foreign Key) – college organizing the fest
- start_date – start date
- end_date – end date
- description – fest details

Relationship

Fest (1) → (N) Events

One fest contains many events.

5. Categories Entity

The Categories table classifies events into different types.

Examples:

- Technical
- Cultural
- Sports
- Hackathon
- Workshop

Attributes

- category_id (Primary Key)
- category_name – type of event

Relationship

Category (1) → (N) Events

One category can contain many events.

6. Events Entity

The Events table is the central entity of the system.

It stores information about each event organized during a fest.

Attributes

- event_id (Primary Key)
- title – name of event
- description – event details
- fest_id (Foreign Key) – fest containing the event
- club_id (Foreign Key) – organizing club
- category_id (Foreign Key) – type of event
- event_date – event date
- venue – location of event
- max_participants – participant limit
- created_at – timestamp

Relationships

Events are connected to:

- registrations
- attendance
- certificates
- teams

Example:

Event → Registrations
Event → Attendance

Event → Certificates
Event → Teams

7. Registrations Entity

The Registrations table records students registering for events.

Attributes

- registration_id (Primary Key)
- user_id (Foreign Key) – student registering
- event_id (Foreign Key) – event registered for
- registration_date – registration timestamp
- status – registration status

Relationship

User (1) → (N) Registrations
Event (1) → (N) Registrations

This forms a many-to-many relationship between users and events.

Example:

Student A → Event1
Student A → Event2
Student B → Event1

8. Attendance Entity

The Attendance table records whether a student attended the event.

Attributes

- attendance_id (Primary Key)
- event_id (Foreign Key)
- user_id (Foreign Key)
- status – present or absent
- marked_at – timestamp

Example:

Event1 → Student1 → Present
Event1 → Student2 → Absent

9. Certificates Entity

The Certificates table stores certificates issued to participants.

Attributes

- certificate_id (Primary Key)
- user_id (Foreign Key) – student receiving certificate
- event_id (Foreign Key) – related event
- certificate_url – link to certificate file
- issued_at – issue date

Example:

Student → Hackathon → certificate.pdf

10. Teams Entity

Some events require team participation, such as hackathons.

Attributes

- team_id (Primary Key)
- event_id (Foreign Key)
- team_name – team name
- created_at – timestamp

Relationship

Event (1) → (N) Teams

One event can have multiple teams.

11. Team Members Entity

The Team Members table stores users belonging to teams.

Attributes

- id (Primary Key)
- team_id (Foreign Key)
- user_id (Foreign Key)

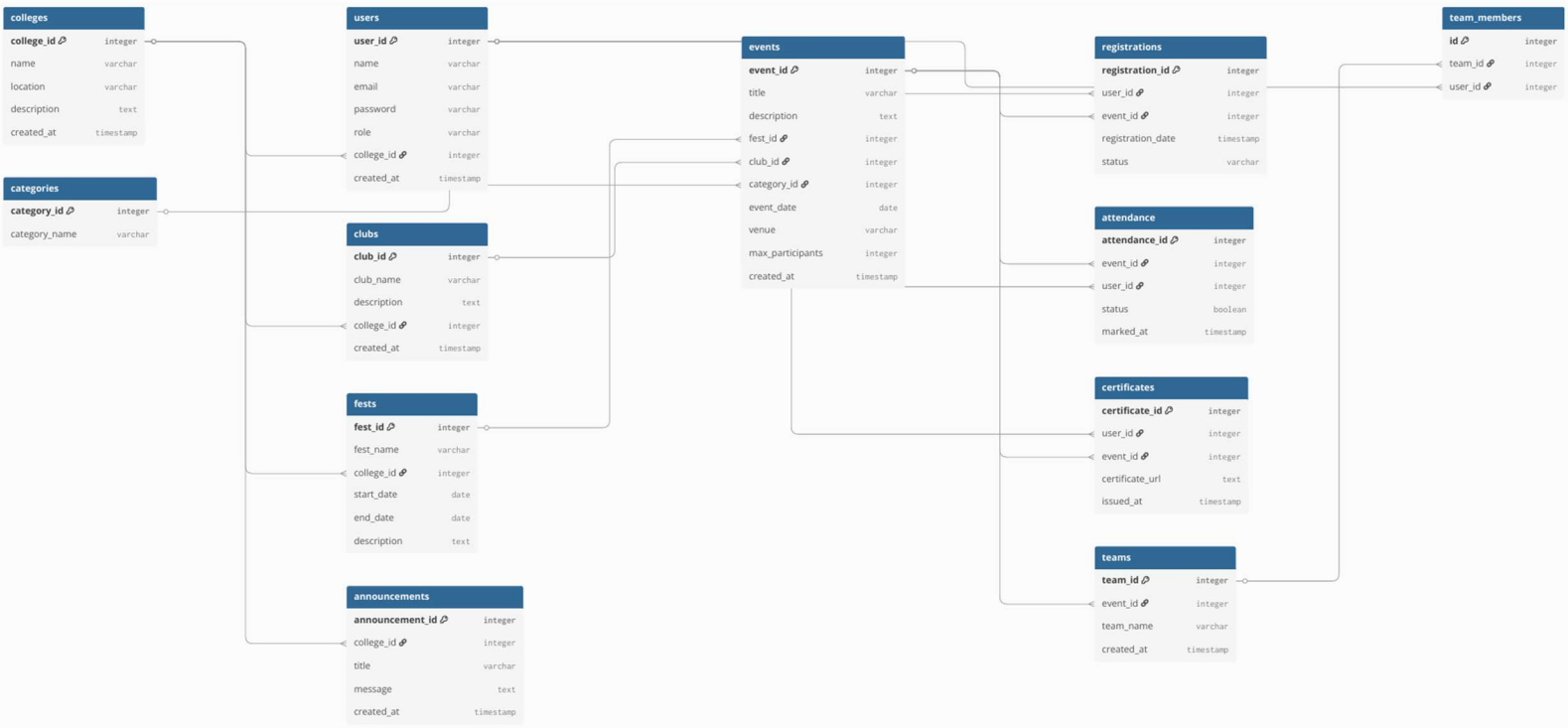
Relationship

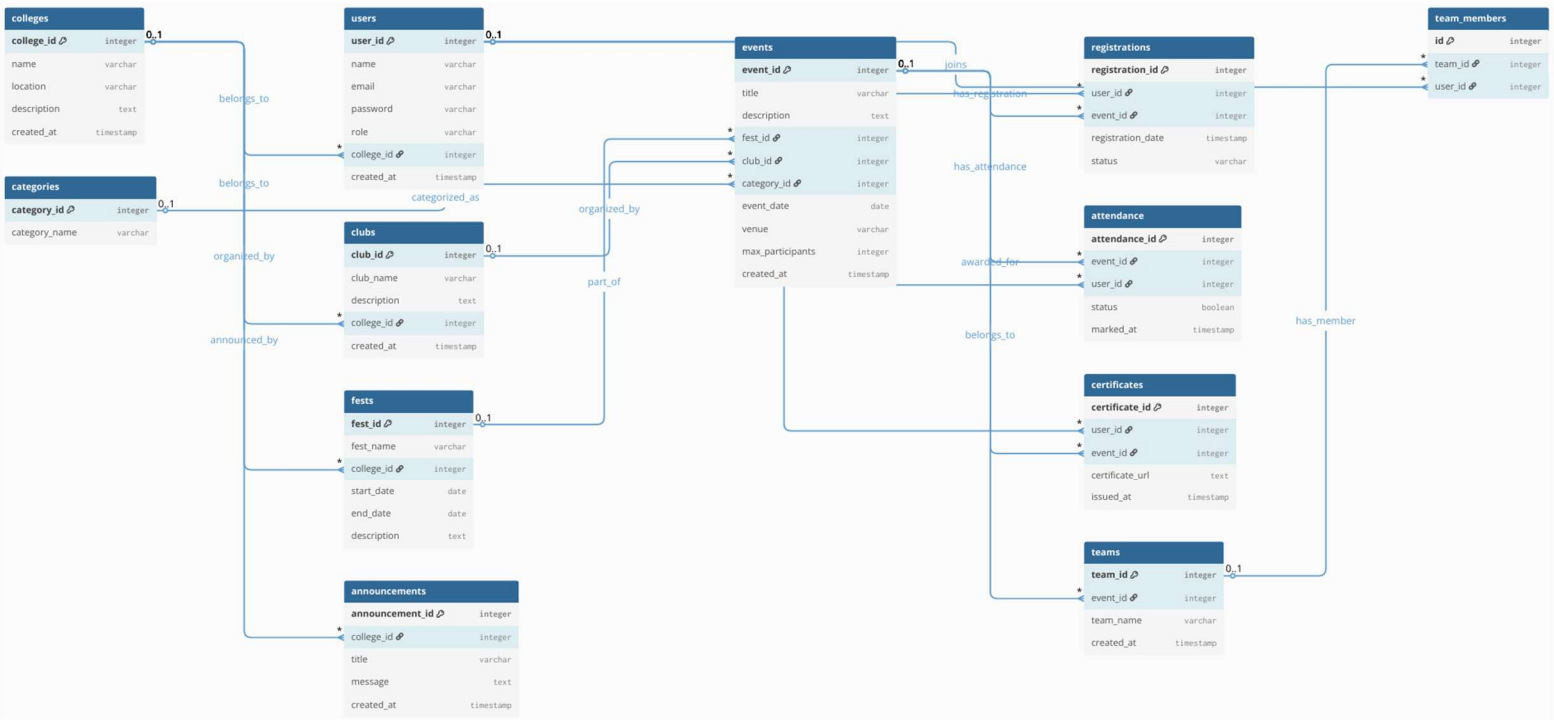
Team (1) → (N) Team Members
User (1) → (N) Team Members

Overall System Flow

The database supports the following workflow:

1. College
2. Clubs organize Events
3. Events belong to Fests
4. Students register for Events
5. Attendance is recorded
6. Certificates are generated




colleges
college_id integer
name varchar
location varchar
description text
created_at timestamp
users
user_id integer
name varchar
email varchar
password varchar
role varchar
college_id integer
created_at timestamp
events
event_id integer
title varchar
description text
fest_id integer
club_id integer
category_id integer
event_date date
venue varchar
max_participants integer
created_at timestamp
registrations
registration_id integer
user_id integer
event_id integer
registration_date timestamp
status varchar
team_members
id integer
team_id integer
user_id integer
categories
category_id integer
category_name varchar
clubs
club_id integer
club_name varchar
description text
college_id integer
created_at timestamp
attendance
attendance_id integer
event_id integer
user_id integer
status boolean
marked_at timestamp
fests
fest_id integer
fest_name varchar
college_id integer
start_date date
end_date date
description text
certificates
certificate_id integer
user_id integer
event_id integer
certificate_url text
issued_at timestamp
announcements
announcement_id integer
college_id integer
title varchar
message text
created_at timestamp
teams
team_id integer
event_id integer
team_name varchar
created_at timestamp
belongs_to
belongs_to
organized_by
announced_by
categorized_as
organized_by
part_of
joins
has_registration
has_attendance
awarded_for
belongs_to
has_member
0..1
*